№13 $\alpha = 0.05$

$\xi \sim N(\theta, a^2)$ $\vec{x}_n$ $n = 139$

$\eta \sim N(\psi, b^2)$ $\vec{y}_m$ $m = 1000$ Незав

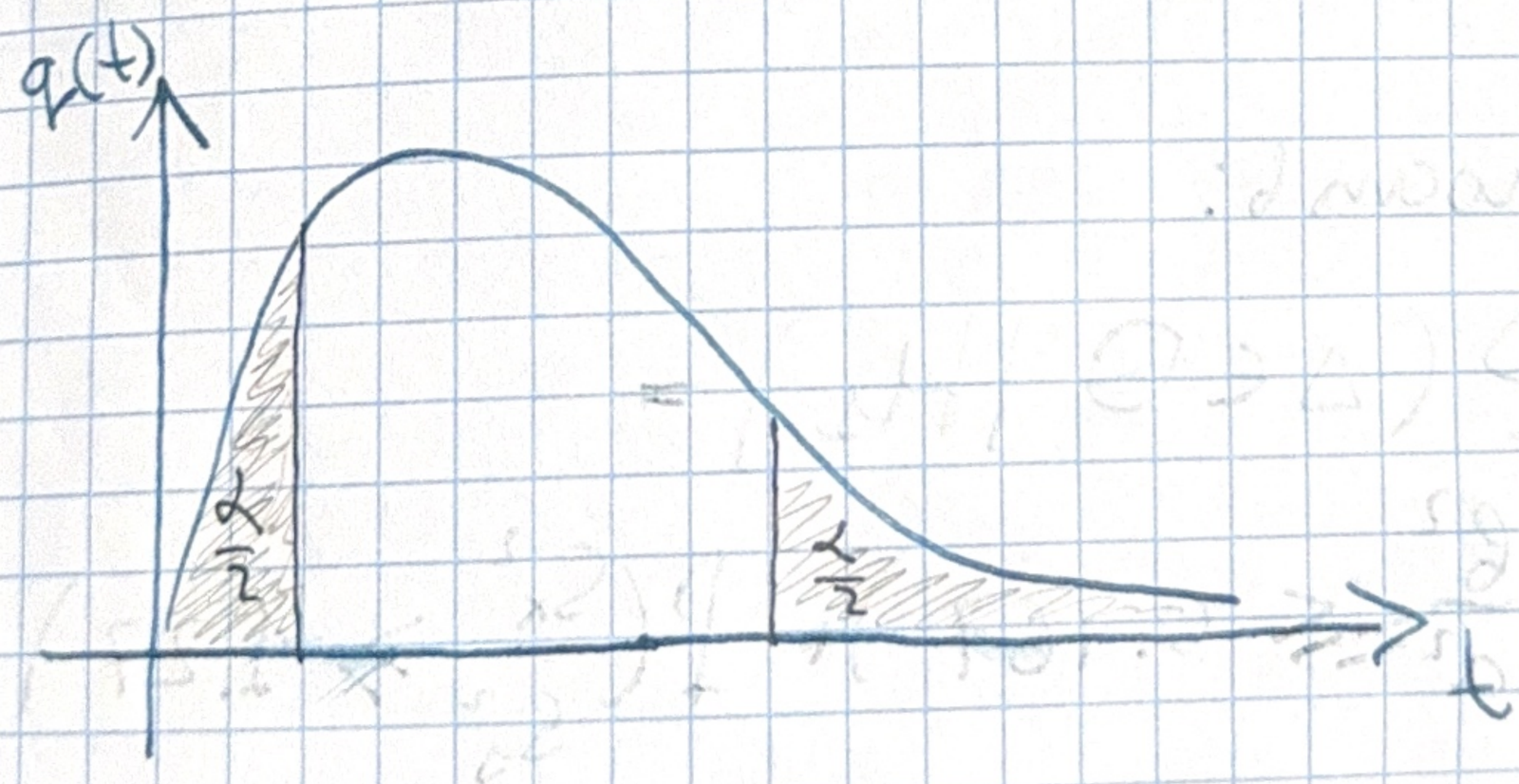
$H_0: a^2 = b^2$

$H_1: a^2 \neq b^2$

$$\frac{(n-1)S_x^2}{a^2} \sim \chi^2(n-1)$$

$$\frac{(m-1)S_y^2}{b^2} \sim \chi^2(m-1)$$

$$\Delta = \frac{S_x^2/a^2}{S_y^2/b^2} \sim F(n-1, m-1)$$



$$P(\Delta \le F_{\frac{\alpha}{2}}(n-1, m-1) \mid H_0) + P(\Delta \ge F_{1-\frac{\alpha}{2}}(n-1, m-1) \mid H_0) = \alpha$$

$$G: \frac{S_x^2}{S_y^2} \le F_{\frac{\alpha}{2}}(n-1, m-1) = 1.27$$

$$\frac{S_x^2}{S_y^2} \ge F_{1-\frac{\alpha}{2}}(n-1, m-1) = 0.767$$

Дима

$$\Delta = \frac{S_x^2}{S_y^2} = 0.863$$

(a=b при H_0)

Нет веских оснований отвергнуть H_0

Мирика

$$\Delta = \frac{S_x^2}{S_y^2} = 0.832$$

Нет веских оснований отвергнуть H_0

Мощность:

$$W = P(\Delta \in G \mid H_1) =$$

$$= P\left(\frac{S_x^2}{S_y^2} \le 0.767\right) + P\left(\frac{S_x^2}{S_y^2} \ge 1.27\right) =$$

$$\frac{S_x^2}{S_y^2} = \Delta \underbrace{\frac{a^2}{b^2}}_{c} = \Delta \cdot c$$

$$P(\Delta \cdot c \leq 0.767) +$$

$$+ P(\Delta \cdot c \geq 1.27) = P\left(\Delta \leq \frac{0.767}{c}\right) + P\left(\Delta \geq \frac{1.27}{c}\right)$$

№ 14

$\xi \sim N(a, 2)$ $\quad x = \{-1.11, -6.10, 2.42\}$

$\eta \sim N(b, 1)$ $\quad y = \{-2.29, -2.91\}$

незав.

$H_0: a = b$

$H_1: a > b$

(T) Фишера:

$$\frac{\bar{x} - a}{\sqrt{2}} \sqrt{3} \sim N(0, 1)$$

$$\frac{\bar{y} - b}{1} \sqrt{2} \sim N(0, 1)$$